

COMP-SCI-5565

Intro to Statistical Learning

Topic 1 – Introduction

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Lecture Objectives

- To read mathematical notation used in this course.
- To compare supervised and unsupervised learning.
- To compare regression and classification problems.
- To understand Parametric vs Nonparametric Models
- To describe the trade-off between more accurate models and more interpretable models.
- To measure the accuracy/goodness of regression model fits.
- To measure the accuracy/goodness of classification model fits.
- To describe how bias and variance contribute to the model error.

Outline

1.1 What is statistical learning?

1.2 Assessing Model Accuracy

1.3 Bias-Variance Trade-off

Outline

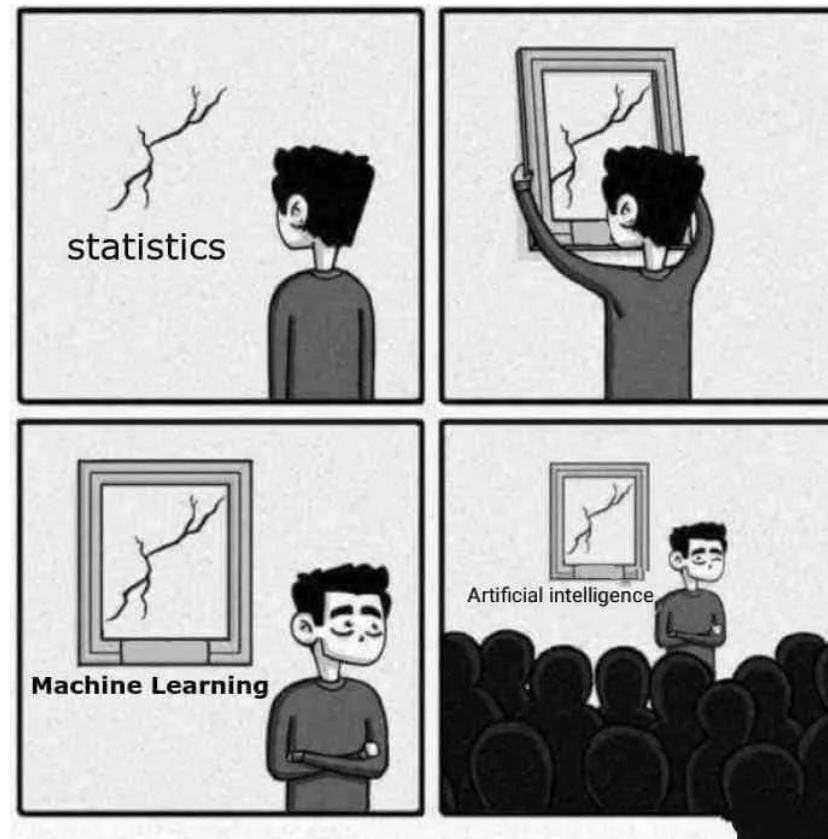
1.1 What is statistical learning?

1.2 Assessing Model Accuracy

1.3 Bias-Variance Trade-off

1.1 What is Statistical Learning?

- Statistical learning is the theoretical foundation for machine learning.
- Statistical learning theory deals with the problem of finding a predictive function based on data—supervised learning.



1.1 What is Statistical Learning?

Some examples of the problems addressed with statistical learning

- Identify the risk factors for some types of cancers.
- Predict whether someone will have a heart attack based on demographic, diet, and clinical measurements.
- Email spam detection.
- Classify a tissue sample into one of several cancer classes based on a gene expression profile.
- Establish the relationship between salary and demographic variables in population survey data.

1.1 What is Statistical Learning?

Types of Statistical Learning Functions

Statistical Learning functions fall into one or more of the following main categories based on how they learn to make predictions or generate content:

- Supervised learning
- Unsupervised learning
- Reinforcement learning
- Generative AI

Other categories are:

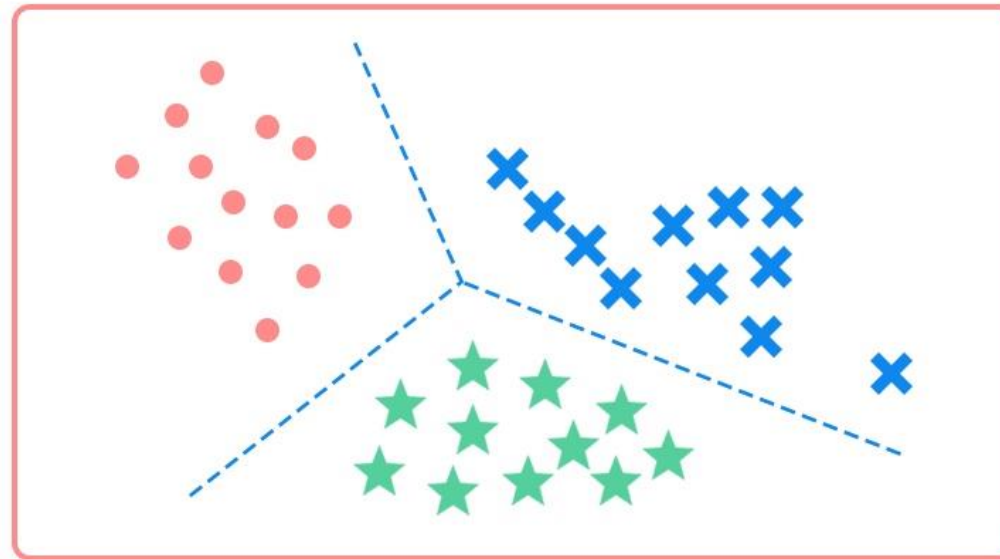
- Semi-supervised learning
- Self-supervised learning

1.1 What is Statistical Learning?

Supervised learning

- Supervised learning: Building a model to predict an output from inputs using labeled data.
 - E.g., Predict market direction from the previous day's performance.

Classification

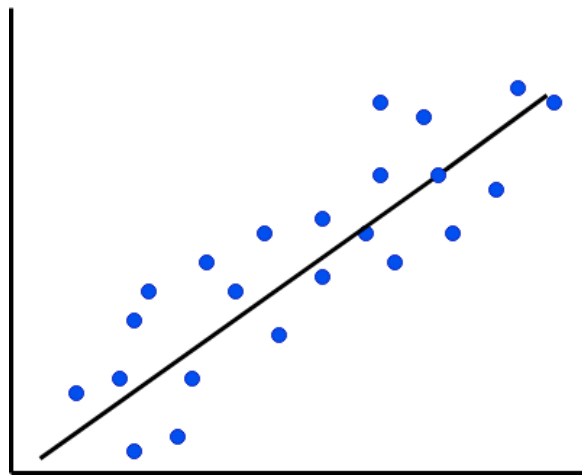


Supervised learning

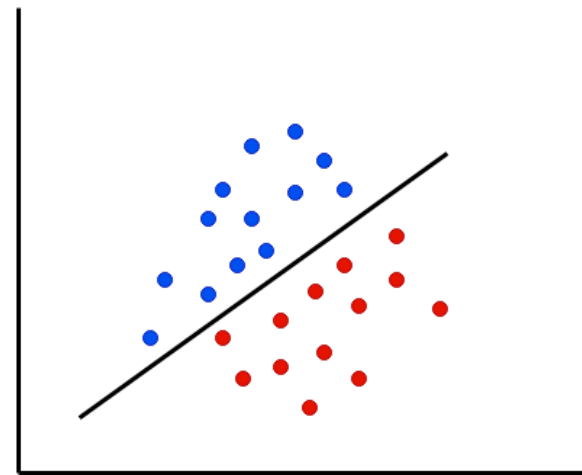
1.1 What is Statistical Learning?

Types of supervised learning problems: Regression vs Classification

- Variables can be characterized as either *quantitative* or *qualitative* (categorical).
- Quantitative variables take on numerical values.
 - E.g., age, sales, temperature, weight, height
- Qualitative variables take on values in *classes*, or categories.
 - E.g., animal, bird species, car make



Regression

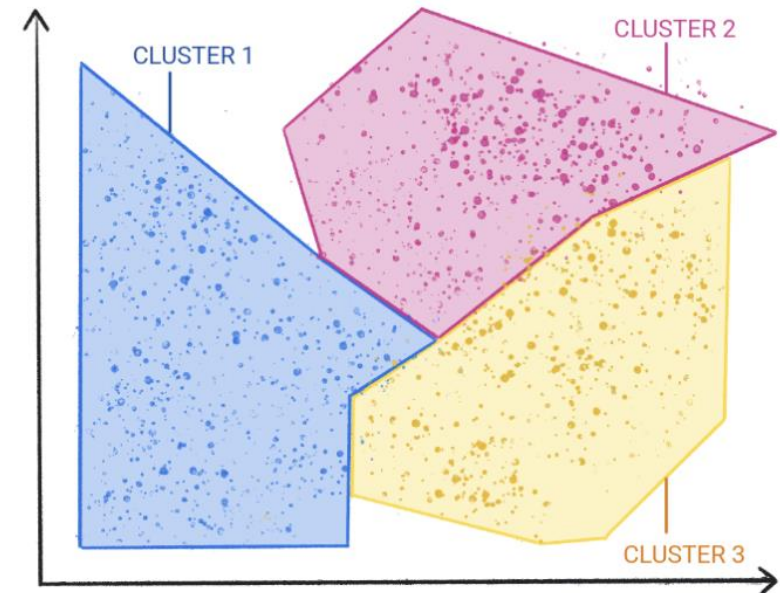
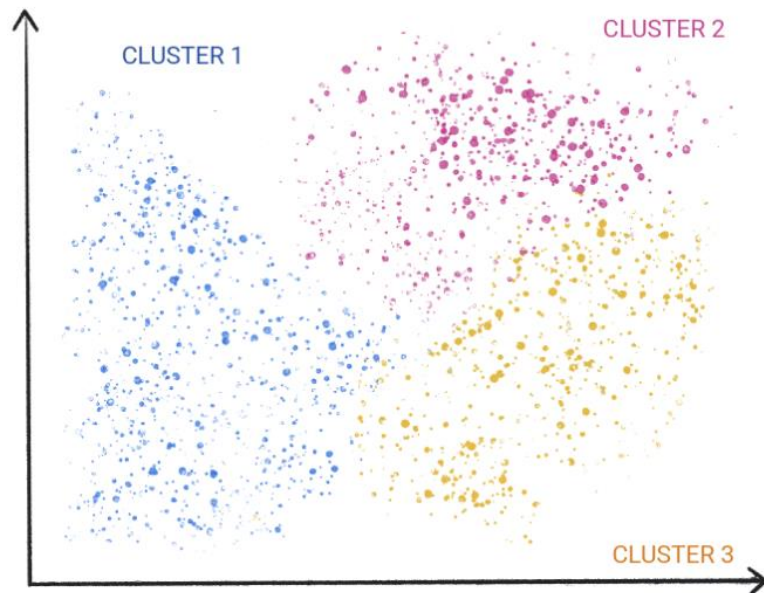


Classification

1.1 What is Statistical Learning?

Unsupervised learning

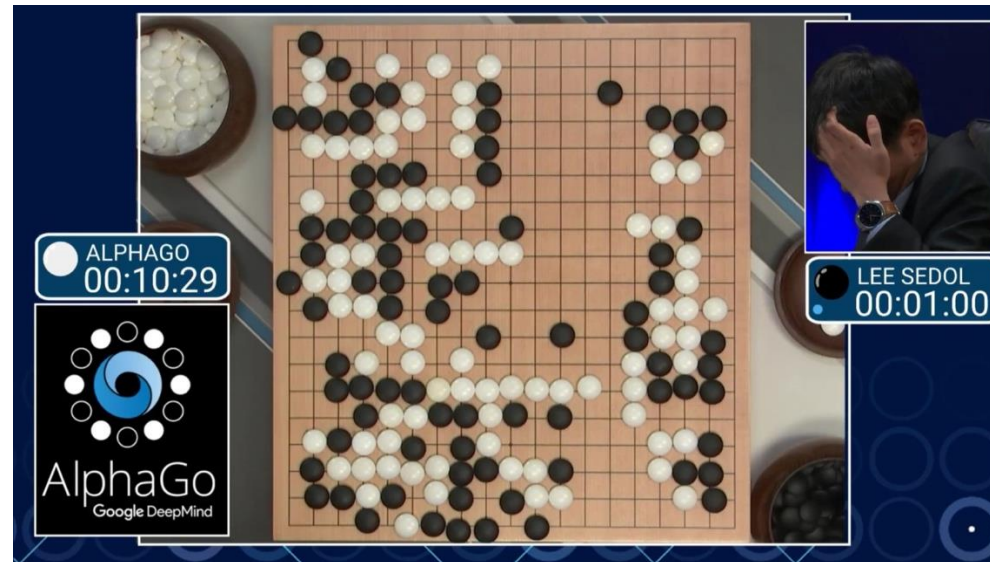
- Unsupervised learning: Inputs but no specific outputs, find relationships and structure.
- Unsupervised learning is commonly used for **clustering problems**.
- E.g., Identify clusters within cancer cell lines.



1.1 What is Statistical Learning?

Reinforcement learning

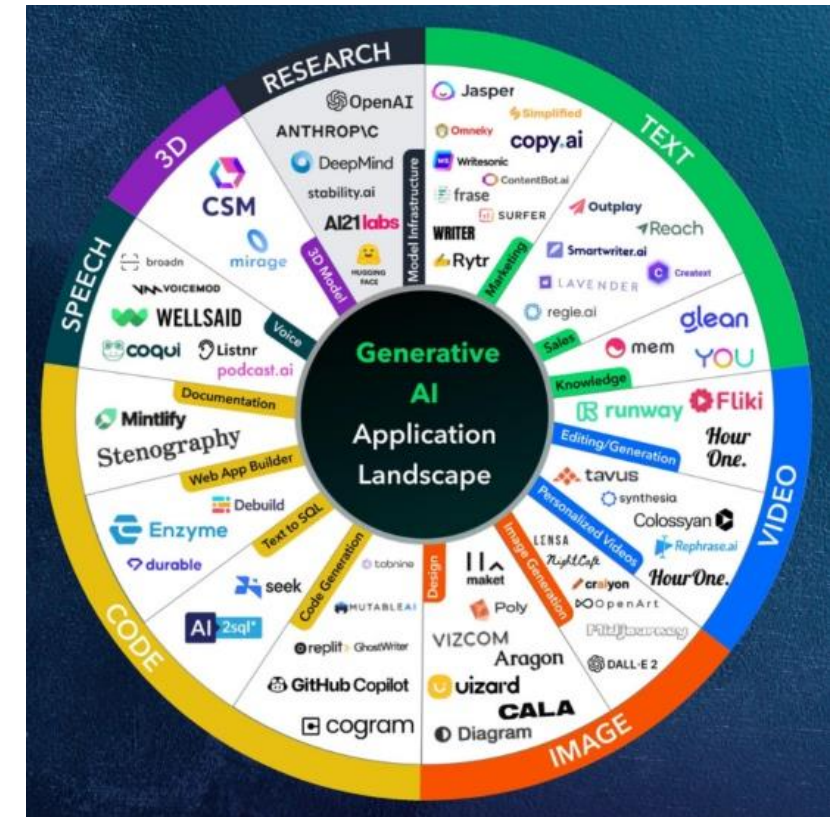
- Reinforcement learning models make predictions by getting **rewards** or penalties based on actions performed within an environment.
- A reinforcement learning system generates a **policy** that defines the best strategy for getting the most rewards.
- AlphaGo - <https://deepmind.google/research/breakthroughs/alphago/>



1.1 What is Statistical Learning?

Generative AI

- Generative AI is a class of models that creates content from user input.
- For example, generative AI can create unique images, music compositions, and jokes; it can summarize articles, explain how to perform a task or edit a photo.
- Generative models by their inputs and outputs:
 - Text-to-text: ChatGPT, Bard, Claude
 - Text-to-image: Imagen
 - Text-to-video: Phenaki
 - Text-to-code: ChatGPT, Bard, Claude
 - Text-to-speech: PlayDialog
 - Image and text-to-image: Google DeepMind



1.1 What is Statistical Learning?

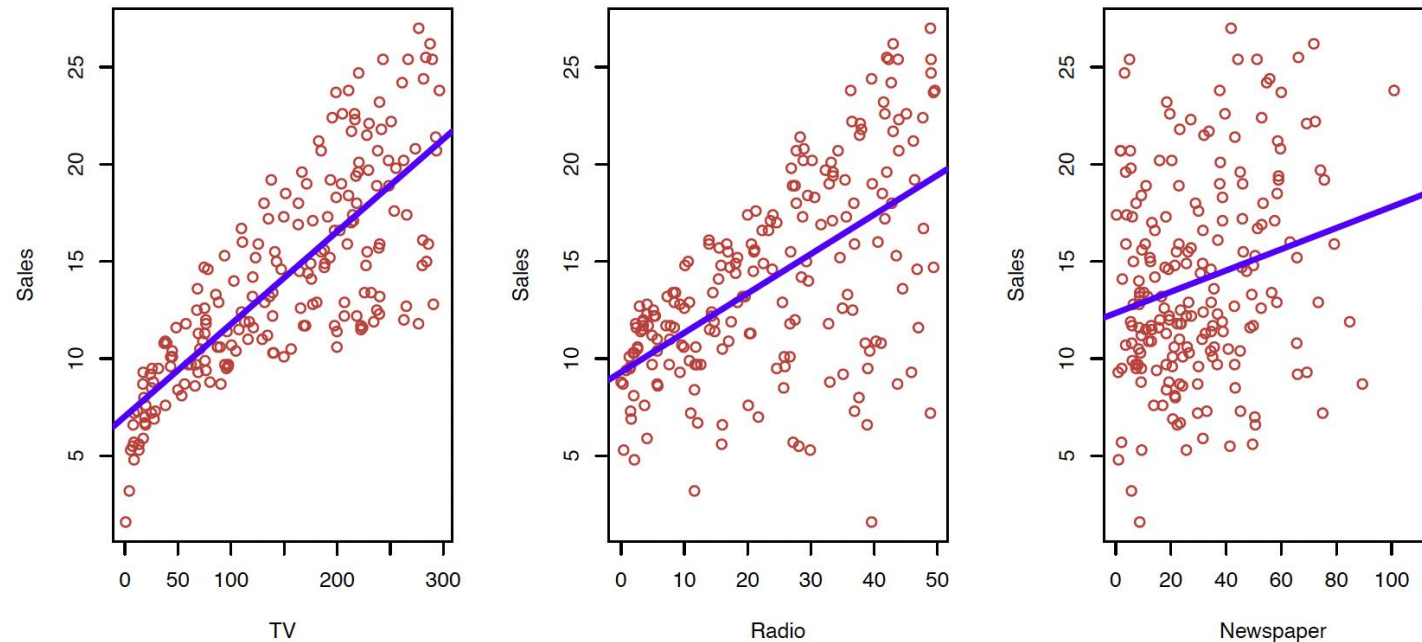
Notations

- n = number of observations/samples (rows)
- p = number of features/variables/predictors (columns)
- y_i = one output/response/label/dependent variable
- $Y = \{y_1, y_2, y_3, \dots y_n\}$ a set of outputs/responses/labels
- x_i = one sample with features/variables/predictors
- $X = \{x_{i1}, \dots x_{np}\}$ a set of samples

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{bmatrix} \begin{matrix} n \text{ samples} \\ p \text{ predictors} \end{matrix}$$

1.1 What is Statistical Learning?

Example problem



- **Sales vs. TV, Radio, and Newspaper** are shown, with a blue linear-regression line fit separately to each.
- Can we predict Sales using these three predictors?
- $Sales \approx f(TV, Radio, Newspaper)$

1.1 What is Statistical Learning?

Example problem – continued

- Sales is a *response* or *target* that we wish to predict. We generically refer to the response as Y , $Y = \text{Sales}$.
- TV is a feature/input/predictor; we name it X_1 .
- Likewise, Radio as X_2 , and so on.

$$X = TV(X_1), Radio(X_2), Newspaper(X_3)$$

- We can refer to the input vector collectively as

$$X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

- We can rewrite the model as

$$Y = f(X) + \varepsilon$$

where f is an unknown function and ε captures measurement errors and other discrepancies (*irreducible error*).

1.1 What is Statistical Learning?

What is $f(X)$ good for?

- With a good f we can make **predictions** of Y at new points $X = x$.
- We can understand which components of $X = \{x_1, x_2, \dots, x_p\}$ are important in explaining Y , and which are irrelevant --- **inference**.
- E.g., TV and Radio may have a big impact on Sales, but Newspaper typically may not.
- Depending on the complexity of f , we may be able to understand how each component x_i of X affects Y .

1.1 What is Statistical Learning?

Why estimate $f(X)$?

- There are two main reasons that we may estimate f : prediction and inference.

Prediction

- In many situations, a set of inputs X are readily available, but the output Y cannot be easily obtained.
- We can predict Y using $\hat{Y} = \hat{f}(X)$.
- $\hat{f}$ is often treated as a black box.
- The accuracy of $\hat{Y}$ as a prediction for Y depends on two quantities, called the *reducible error* and the *irreducible error*.

1.1 What is Statistical Learning?

Why estimate $f(X)$? – continued

- *Reducible error:* We can reduce the gap between our estimate and the true function by applying improved methods. f vs $\hat{f}$
- *Irreducible Error:* The quantity ε may contain unmeasured variables that are useful in predicting Y : since we don't measure them, $\hat{f}$ cannot use them for its prediction. The quantity ε will also contain unmeasurable variation.



1.1 What is Statistical Learning?

Why estimate $f(X)$? – continued

Inference

- Understanding the association between Y and $X = \{x_1, x_2, \dots, x_p\}$.
- We wish to estimate f , but our goal is not necessarily to make predictions for Y .
- Now f cannot be treated as a black box, because we need to know its exact form.
- One may be interested in answering the following questions:
 - Which media are associated with sales?
 - Which media generate the biggest boost in sales? Or
 - How large of an increase in sales is associated with a given increase in TV advertising?

1.1 What is Statistical Learning?

How do we estimate $f(X)$?

- We want to find a function f such that $Y \approx f(X)$ for any observation.
- Most statistical learning methods for this task can be characterized as either *parametric* or *non-parametric*.

Parametric Methods

- Parametric methods involve a two-step model-based approach.
- First, we assume the functional form of f .

$$\text{Assume } f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

- After a form is selected, we apply a procedure to fit or train the model. Estimate $(p + 1)$ parameters, not f .

$$\hat{Y} \approx \hat{\beta}_0 + \hat{\beta}_1 X + \cdots + \hat{\beta}_p X$$

- Disadvantage: model we choose will usually not match the true unknown form of f . If the chosen model is too far from the true f , then our estimate will be poor – high reducible error.

1.1 What is Statistical Learning?

How do we estimate $f(X)$? – continued

- Parametric methods include Linear Regression, Logistic Regression, Lasso, Ridge, Linear Discriminant Analysis, and Quadratic Discriminant Analysis.
- They are less flexible in fitting the data.
- They have high interpretation (inference).

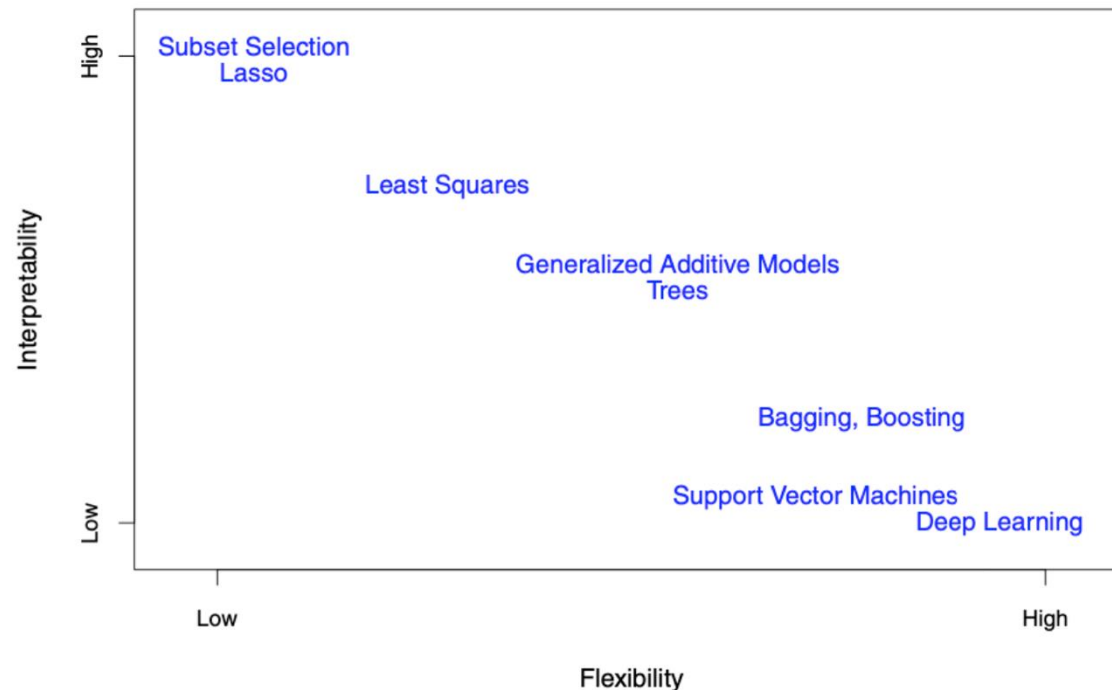
Non-Parametric Methods

- Non-parametric methods don't make explicit assumptions about the functional form of f .
- Instead, they seek an estimate $\hat{f}$ that gets as close to the data points as possible without being too rough or wiggly.
- Disadvantage: A very large number of observations (far more than is typically needed for a parametric approach) is required to obtain an accurate estimate for f .

1.1 What is Statistical Learning?

How do we estimate $f(X)$? – continued

- Non-parametric methods include Bagging, Boosting, Support Vector Machines, and Deep Learning.
- They are more flexible in fitting the data.
- They have medium-to-low interpretation (inference).



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1.3 Bias-Variance Trade-off

1.2 Assessing Model Accuracy

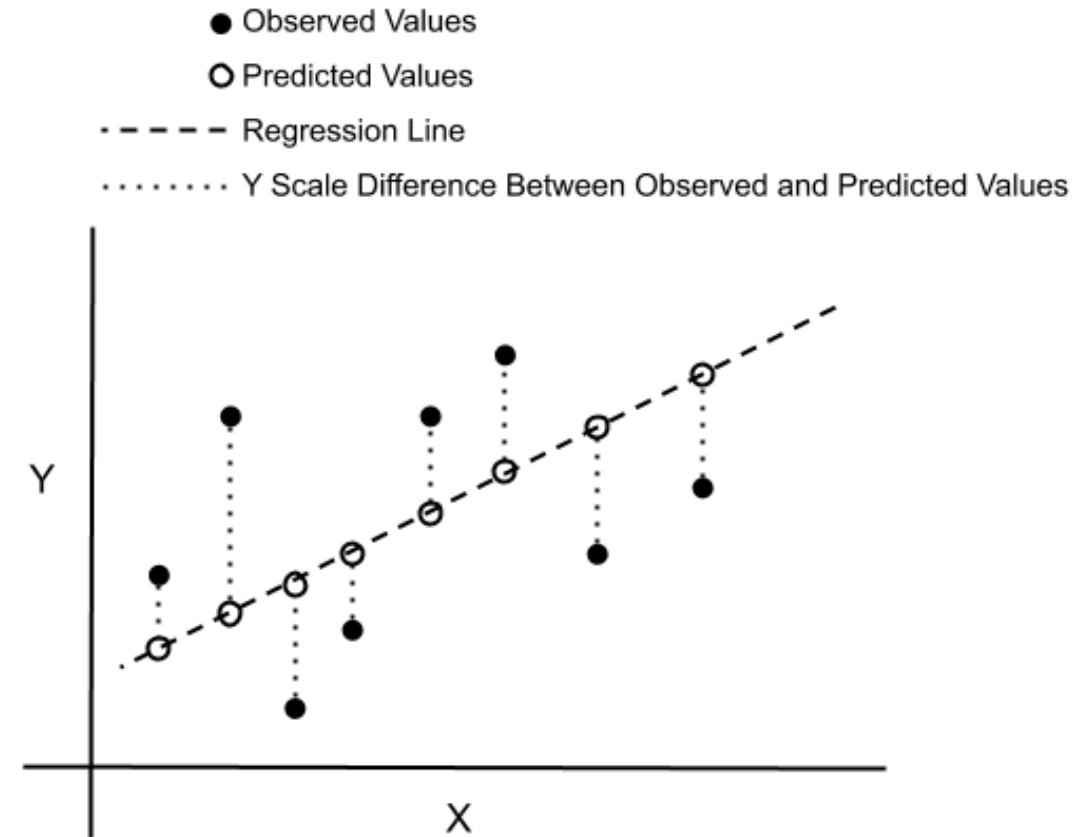
Measuring Quality of Fit

- Quality of fit: Quantify how close the predicted response is to the true response.
- We train with training samples and test with testing samples.

Mean Squared Error (MSE)

- In the regression setting, the most used measure is the *mean squared error (MSE)*, given by

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2,$$



1.2 Assessing Model Accuracy

Mean Squared Error (MSE) – continued

- The squared cancels negatives and amplifies the errors.
- The MSE will be small if the predicted responses are very close to the true responses and large if the predicted and true responses differ substantially.
- The MSE computed using the training data $Tr = \{x_i, y_i\}_1^N$ should more accurately be referred to as the *training MSE*,

$$MSE_{Tr} = Ave_{i \in Tr} (y_0 - \hat{f}(x_0))^2.$$

- We are interested in the accuracy of our predictions when we apply our method to previously unseen *test data* $Te = \{x_i, y_i\}_1^M$.

$$MSE_{Te} = Ave_{i \in Te} (y_0 - \hat{f}(x_0))^2,$$

- Other evaluation metrics include Root-Mean-Squared-Error (RMSE), Mean-Absolute-Error (MAE), Mean-Squared-Percentage-Error (MSPE).

1.2 Assessing Model Accuracy

The Classification Setting

- The most common approach for quantifying the accuracy of our estimate $\hat{f}$ is the training *error rate*, the proportion of mistakes that are made if we apply our estimate $\hat{f}$ to the training observations:

$$\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$$

- The *test error* rate associated with a set of test observations of the form is given by

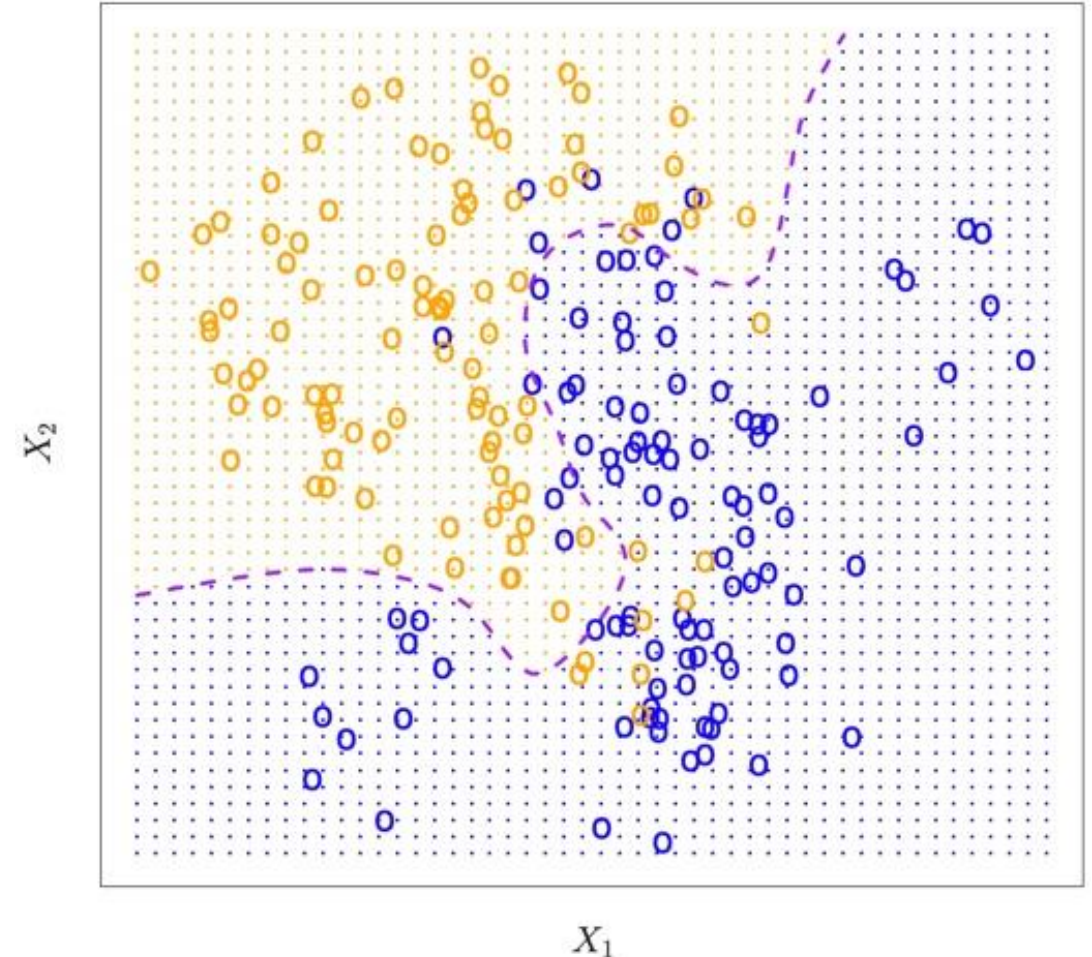
$$Ave(I(y_0 \neq \hat{y}_0))$$

- The test error rate is minimized, on average, by a classifier that *assigns each observation to the most likely class, given its predictor values*.
- We should simply assign a test observation with predictor vector x_0 to the class j for which $\Pr(Y = j \mid X = x_0)$ is the largest – *Bayes Classifier*.

1.2 Assessing Model Accuracy

The Classification Setting – continued

- The purple dashed line represents the points where the probability is exactly 50%.
- An observation that falls on the orange side of the boundary will be assigned to the orange class, and similarly, an observation on the blue side of the boundary will be assigned to the blue class.
- This is known as the Bayes decision boundary
- *Hypothetical – cannot be done in practice.*



Outline

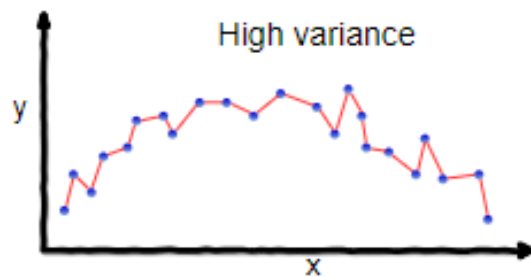
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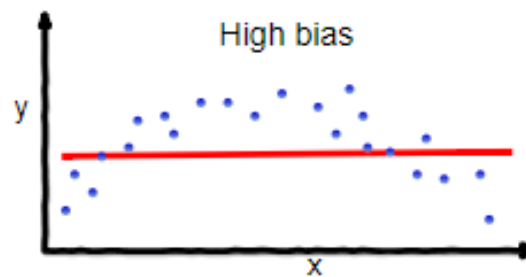
1.3 Bias-Variance Trade-off

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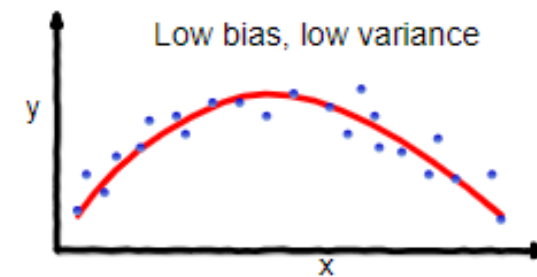
- The *bias-variance tradeoff* describes the relationship between a model's complexity, the accuracy of its predictions, and how well it can make predictions on test data.
- A more flexible model can better fit training data. It is said to have lower error or bias.
- However, more flexible models tend to have greater *variance* in prediction each time we take a set of test samples.



overfitting



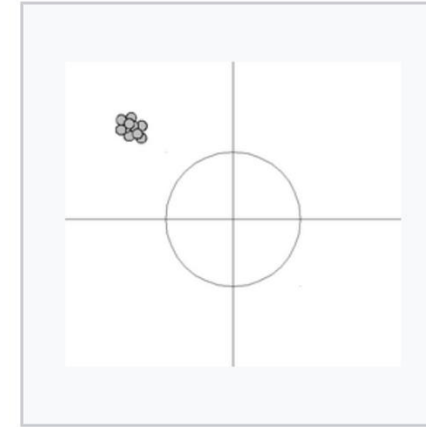
underfitting



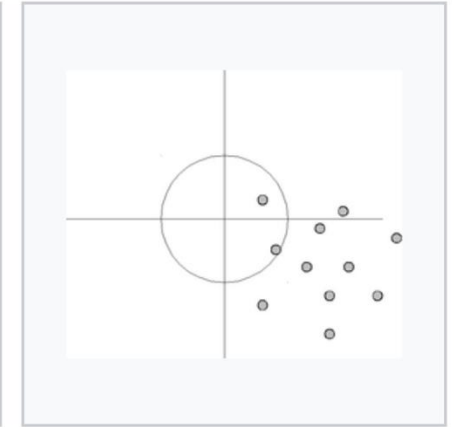
Good balance

1.3 Bias-Variance Trade-off

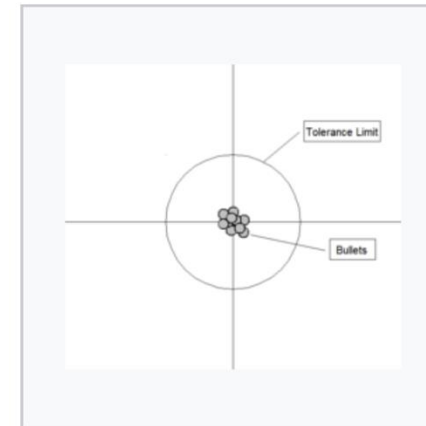
- If an algorithm is too simple, it may have high bias and low variance and thus is error-prone.
- If an algorithm fits too complex, it may have high variance and low bias.
- An algorithm can't be more complex and less complex at the same time.
- We must find a 'middle ground.'



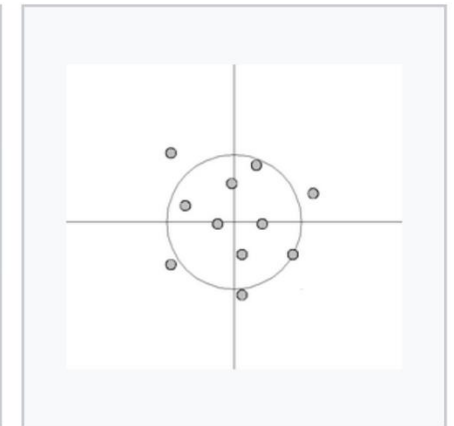
High bias, low variance



High bias, high variance



Low bias, low variance

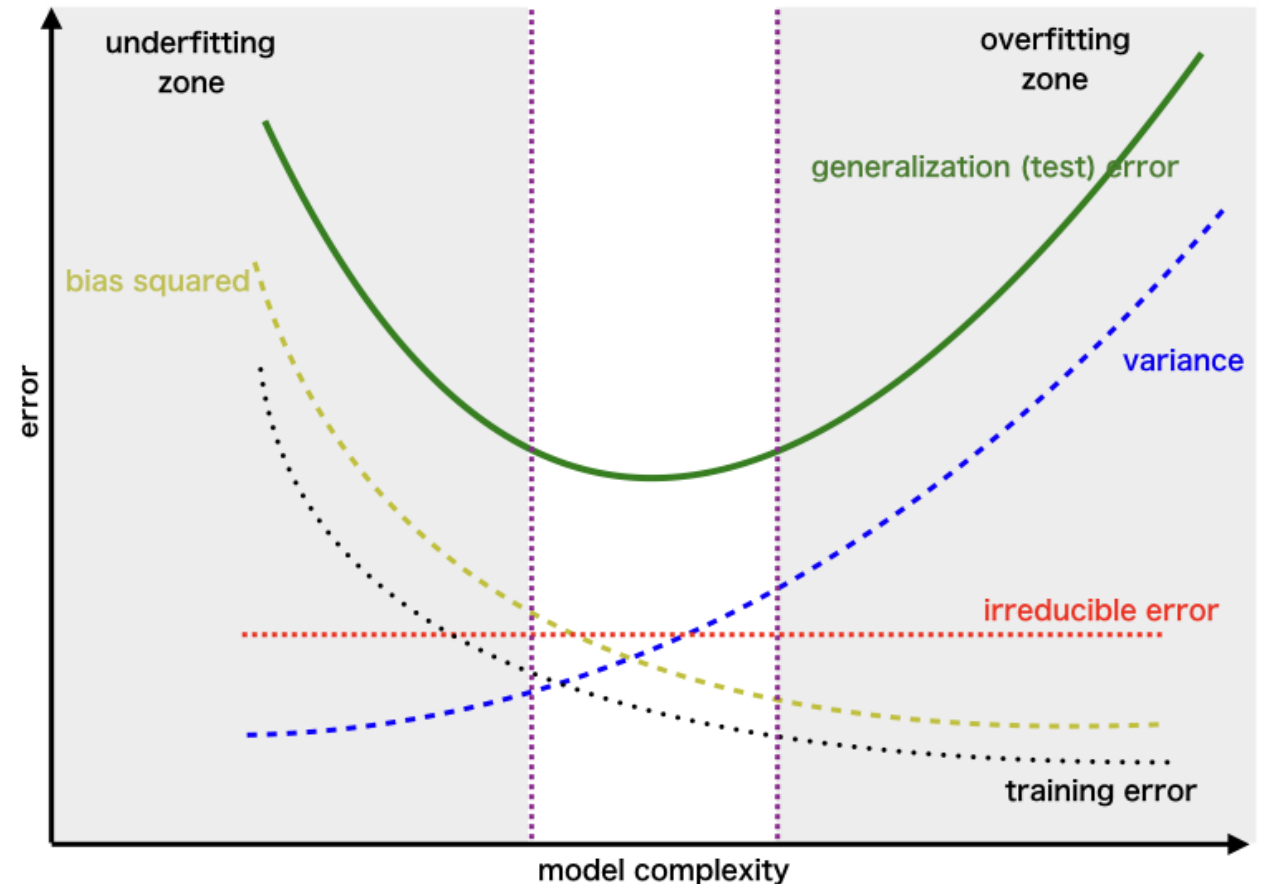


Low bias, high variance

1.3 Bias-Variance Trade-off

Overfitting, Underfitting, and Optimal Fit

- Overfitting: A model with low bias and high variance. Performs well on train data but badly on test data.
- Underfitting: A model with high bias and low variance. Performs poorly on both train and test data.
- Optimal fit: A model that finds a balance between bias and variance. Performs 'good enough' on both train and test data.



Further Reading

- Chapter 1 & 2 – James G., Witten D., Hastie T., Tibshirani R., & Taylor J. (2023) *An Introduction to Statistical Learning with Applications in Python*. Springer
<https://link.springer.com/book/10.1007/978-3-031-38747-0>
- Chapter 1 & 2 - Introduction to Statistical Learning Using R Book Club: <https://r4ds.github.io/bookclub-islr/>
- Bias-Variance Trade: <https://www.geeksforgeeks.org/ml-bias-variance-trade-off/>

Course Resources

- Textbook: <https://link.springer.com/book/10.1007/978-3-031-38747-0>
- Textbook Club: <https://r4ds.github.io/bookclub-islr/>
- Coding environment – Python
 - Local setup – Anaconda
 - Windows & Linux – <https://www.youtube.com/watch?v=MUZtVEDKXsk>
 - Mac – <https://www.youtube.com/watch?v=RFeln2ywxG4>
 - Cloud: Google Colab – <https://www.youtube.com/watch?v=inN8seMm7UI>
- Packages: ISLP, Sklearn, Statsmodel, Numpy, Pandas, Jupyter Notebook
- Datasets: <https://islp.readthedocs.io/en/latest/data.html>

End of Topic

Thank you
Any questions?